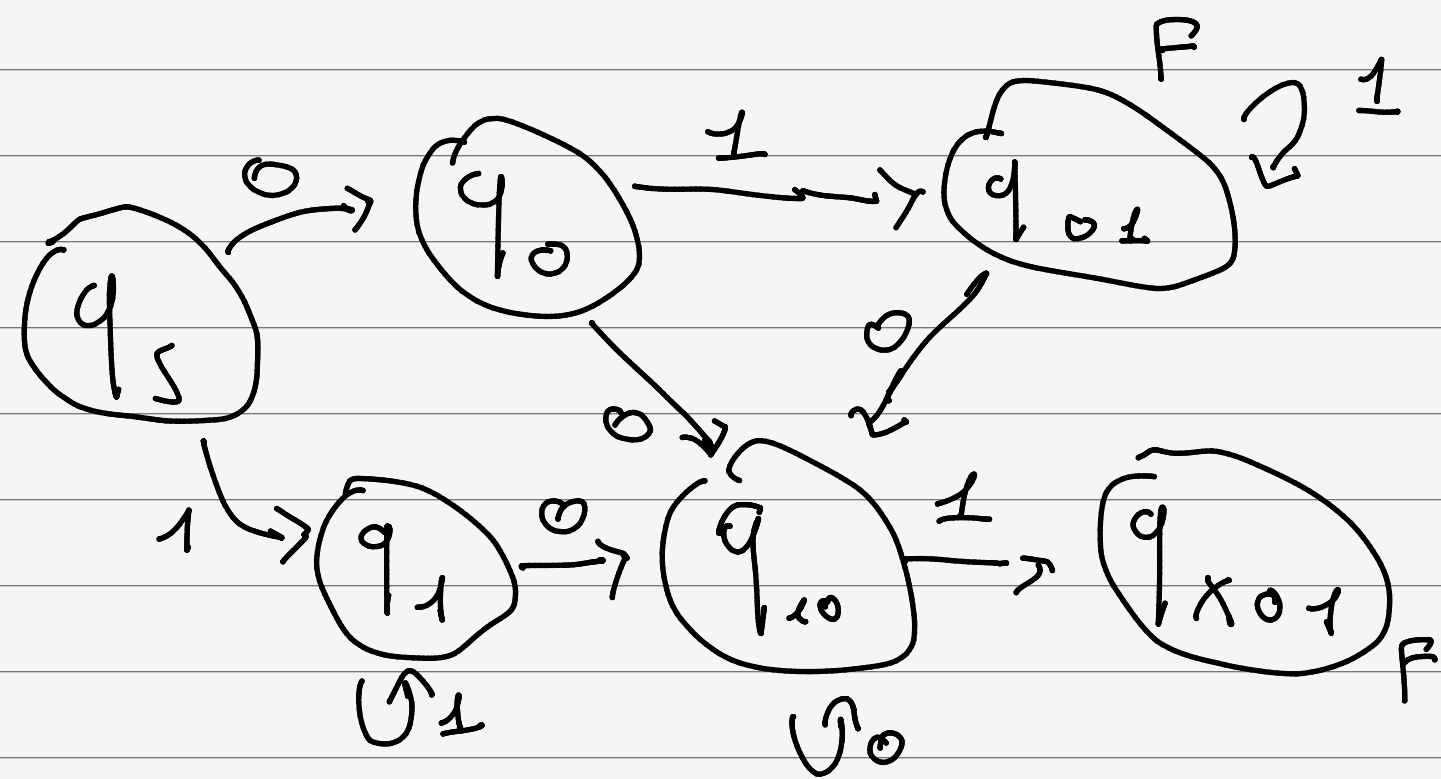
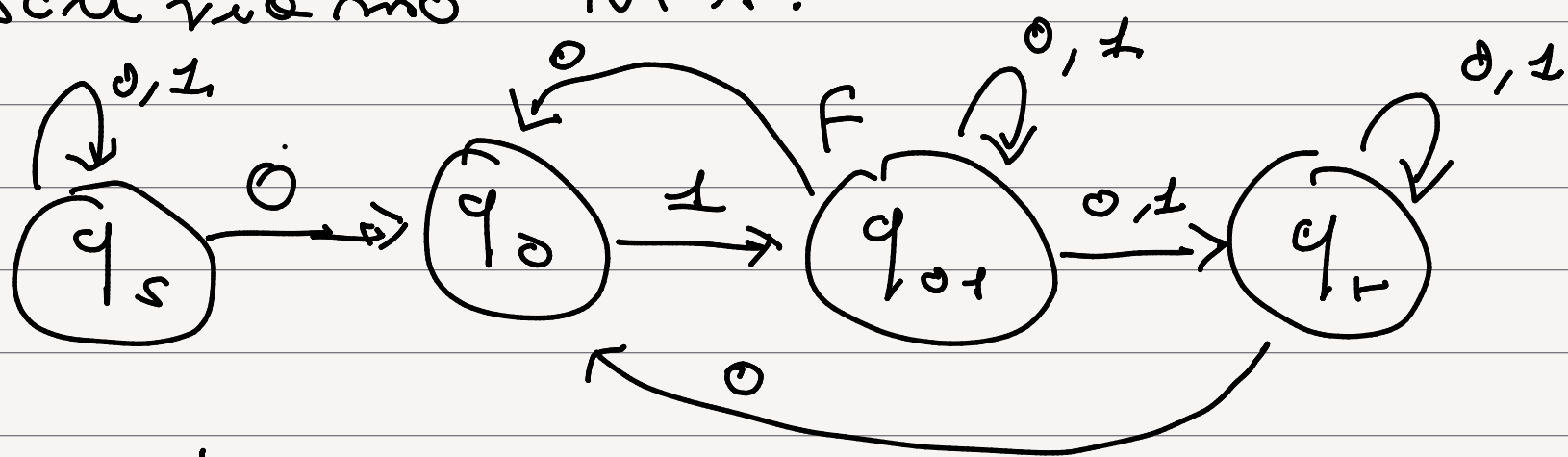


Esercizi su DFA, NFA

$$1) \{w \in \{0,1\}^+ \mid w = 01x = x01 = 01x01, x \in \{0,1\}^*\}$$

Scriviamo NFA:



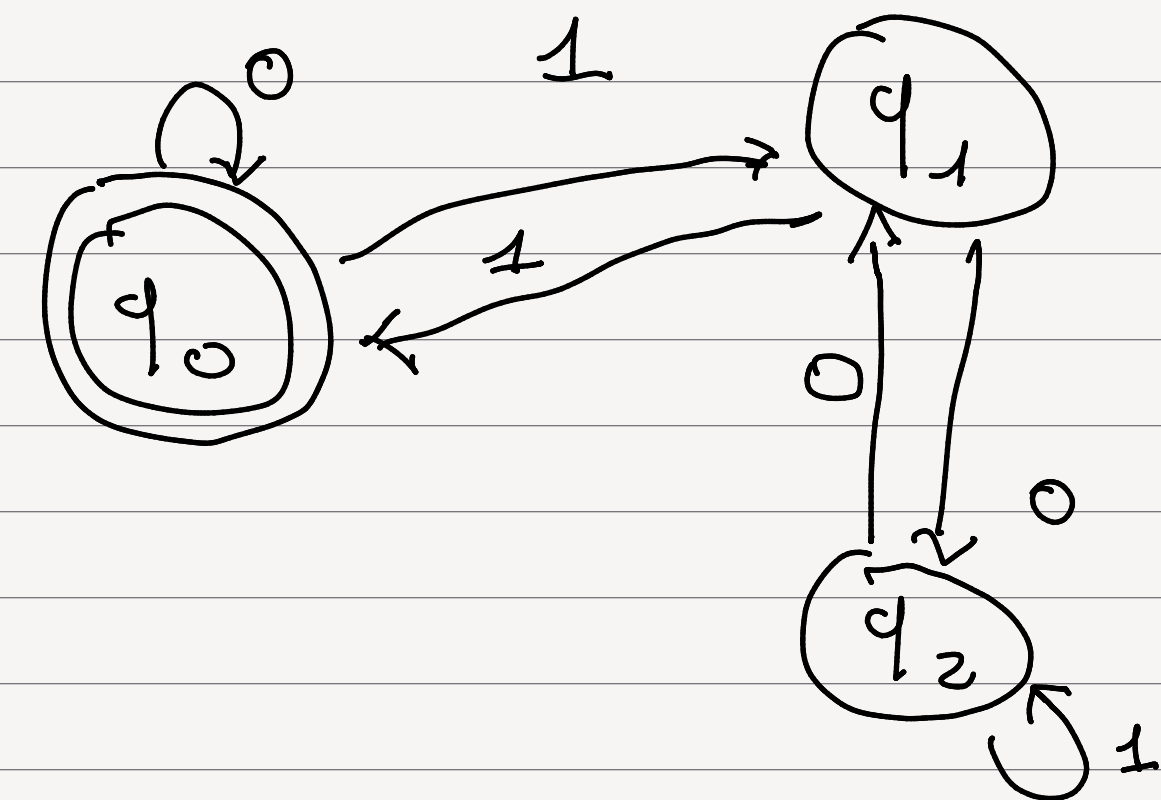
~

~

~

~

$$\{w \in \{0,1\}^* \mid w \text{ multiplo di } 3\}$$



~

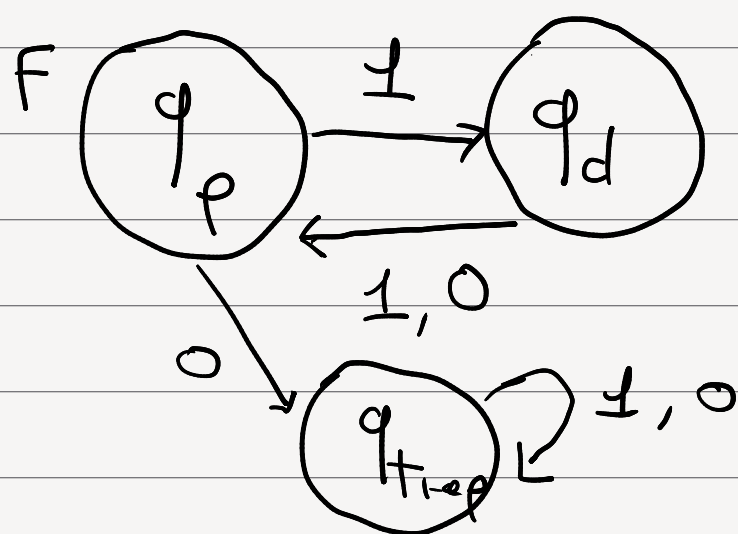
~

~

~

~

$$\{w \in \{0,1\}^* \mid w(i)=1 \quad \forall i \text{ dispari}\}$$

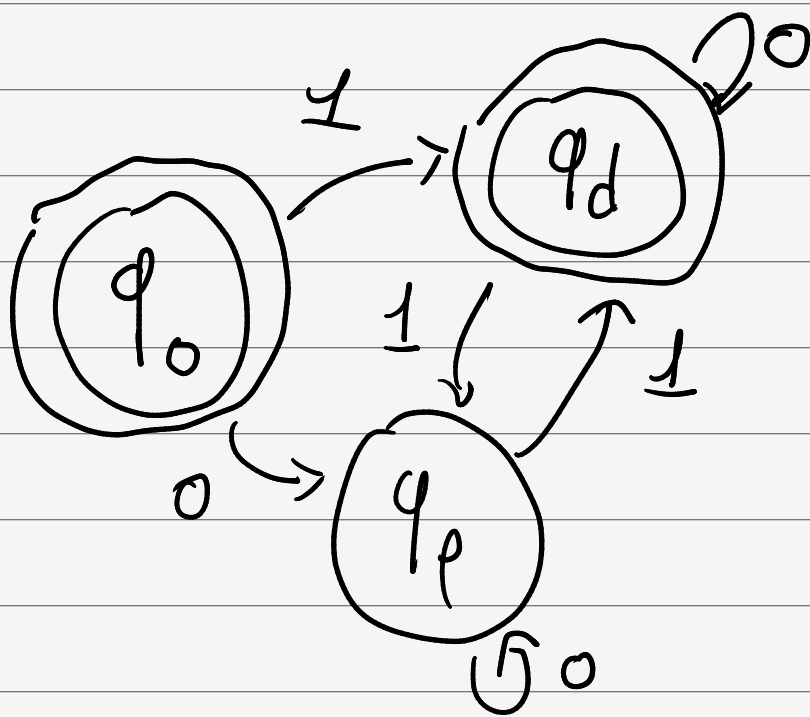


$$S \rightarrow \epsilon \mid 1D$$

$$D \rightarrow 1S \mid 0S \mid 110$$

~ ~ ~ ~ ~

$$\{ w \in \{0,1\}^* \mid \#(1) = 2k+1, k \in \mathbb{N} \}$$



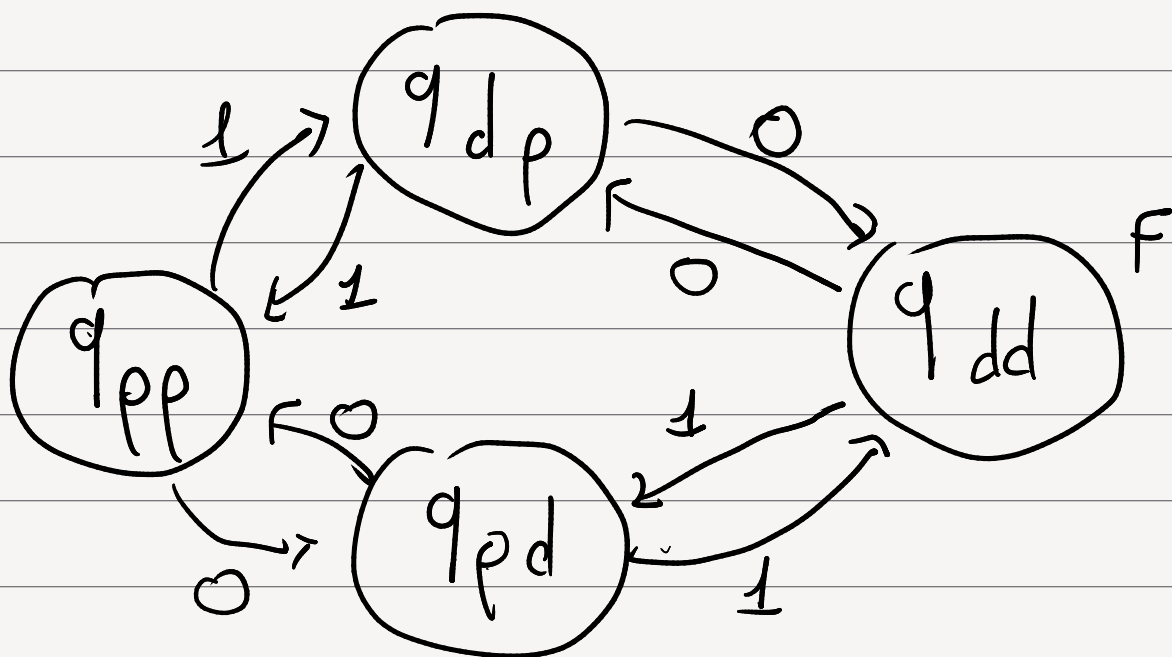
$$S \rightarrow 1D \mid 0P \mid 1 \mid \epsilon$$

$$D \rightarrow 0D \mid 1P \mid 0$$

$$P \rightarrow 0P \mid 1D \mid 1$$

~ ~ ~ ~ ~

$$\{ w \in \{0,1\}^+ \mid \#(0) = 2k+1, \#(1) = 2m+1, k, m \in \mathbb{N} \}$$



$$S \rightarrow 1A \mid 0B$$

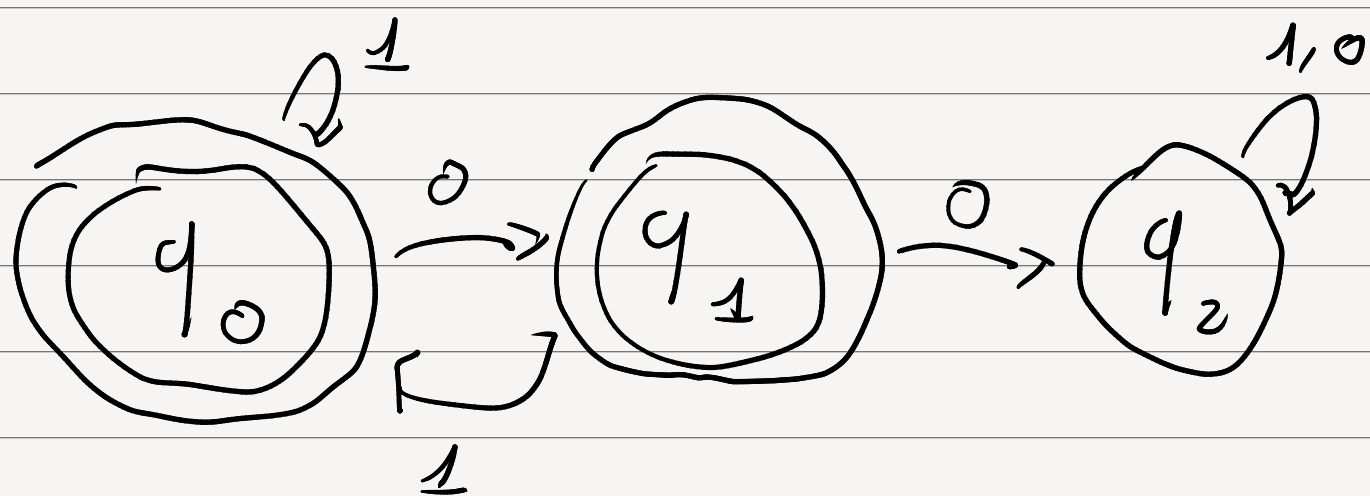
$$A \rightarrow 1S \mid 0D \mid 0$$

$$B \rightarrow 0S \mid 1D \mid 1$$

$$D \rightarrow 0A \mid 1B$$

$\sim$ $\sim$ $\sim$ $\sim$ $\sim$

$\{w \in \{0,1\}^* \mid w \text{ non contiene due } 0 \text{ consecutivi}\}$



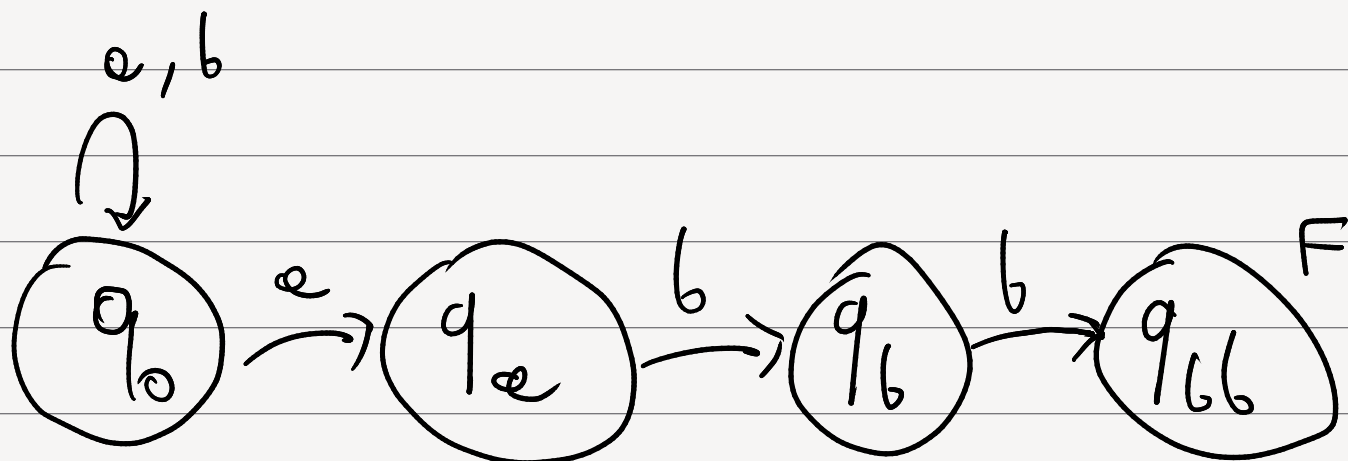
$S \rightarrow 1S / 0A / 1 / 0 / \epsilon$

$A \rightarrow 1S / 1$

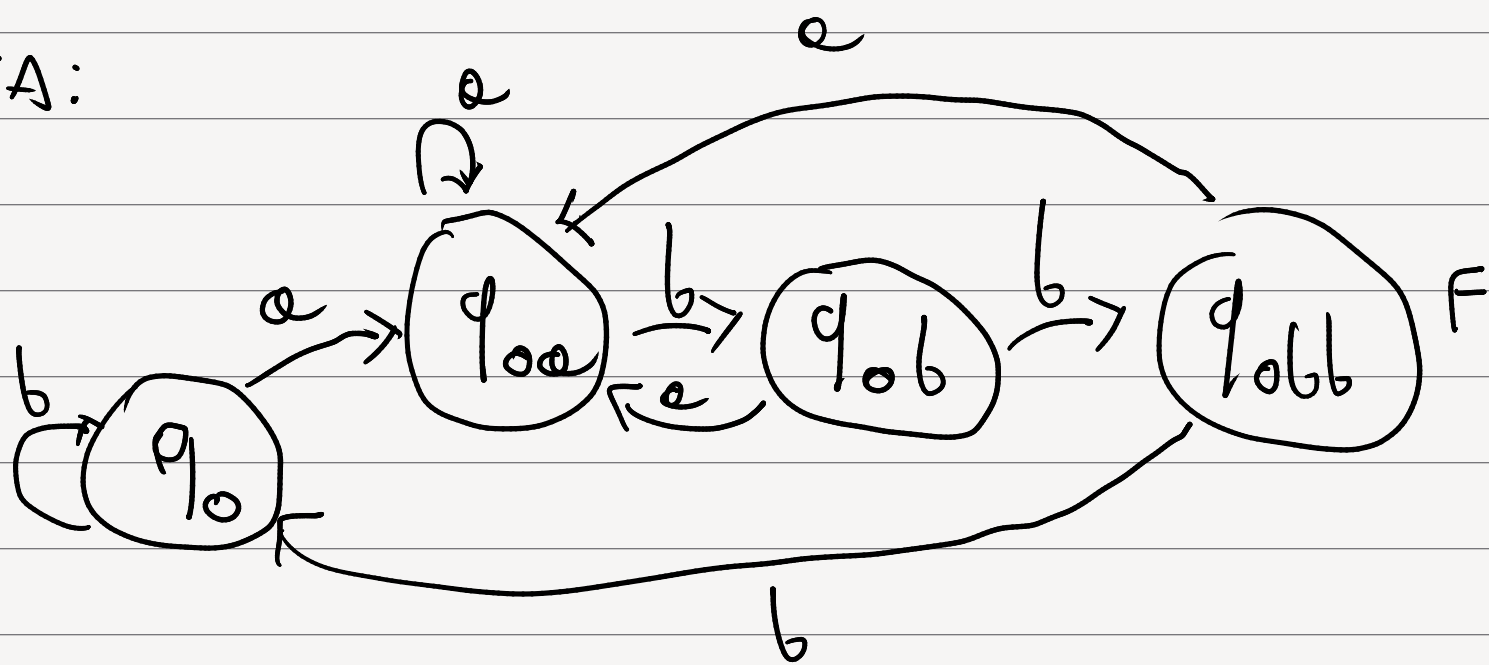
$\sim$ $\sim$ $\sim$ $\sim$

$\{w \in \{a,b\}^* \mid w = (a|b)^* abb\}$

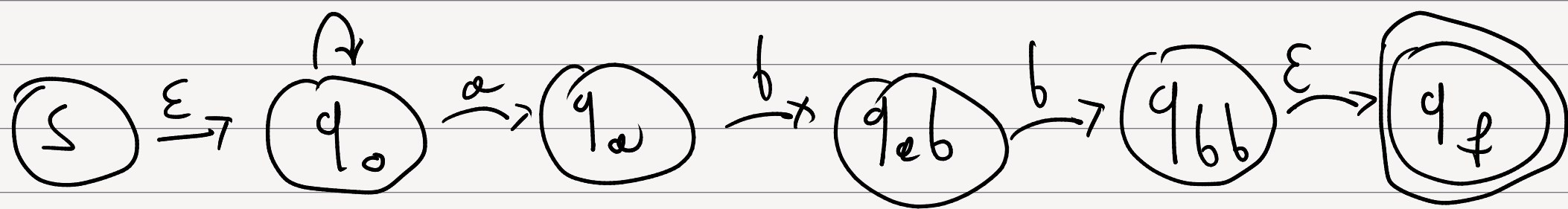
NFA:



DFA:



trasformo l'NFA in GNFA



Sia $q_{rip} = q_0$

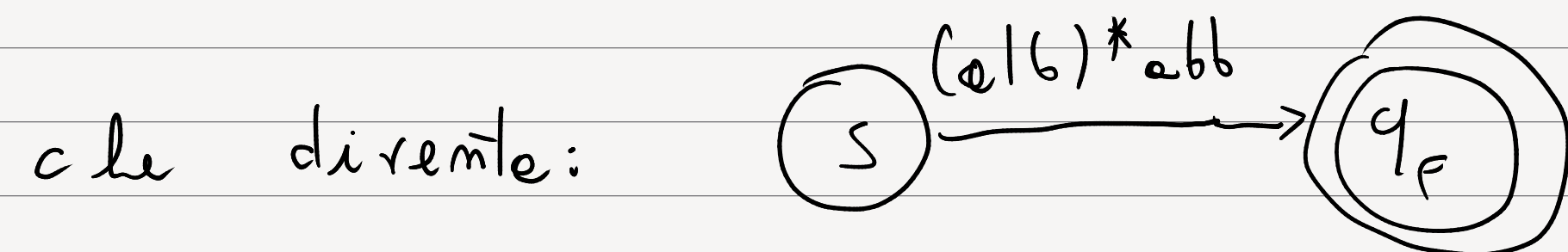
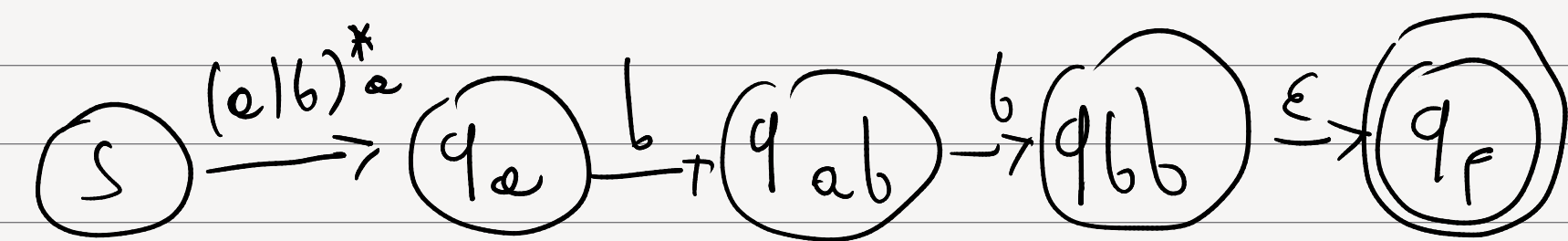
$\delta(q_0, q_0) = (a|b)^*$

$$\delta(s, q_0) = \varepsilon$$

$$\delta(s, q_e) = \emptyset$$

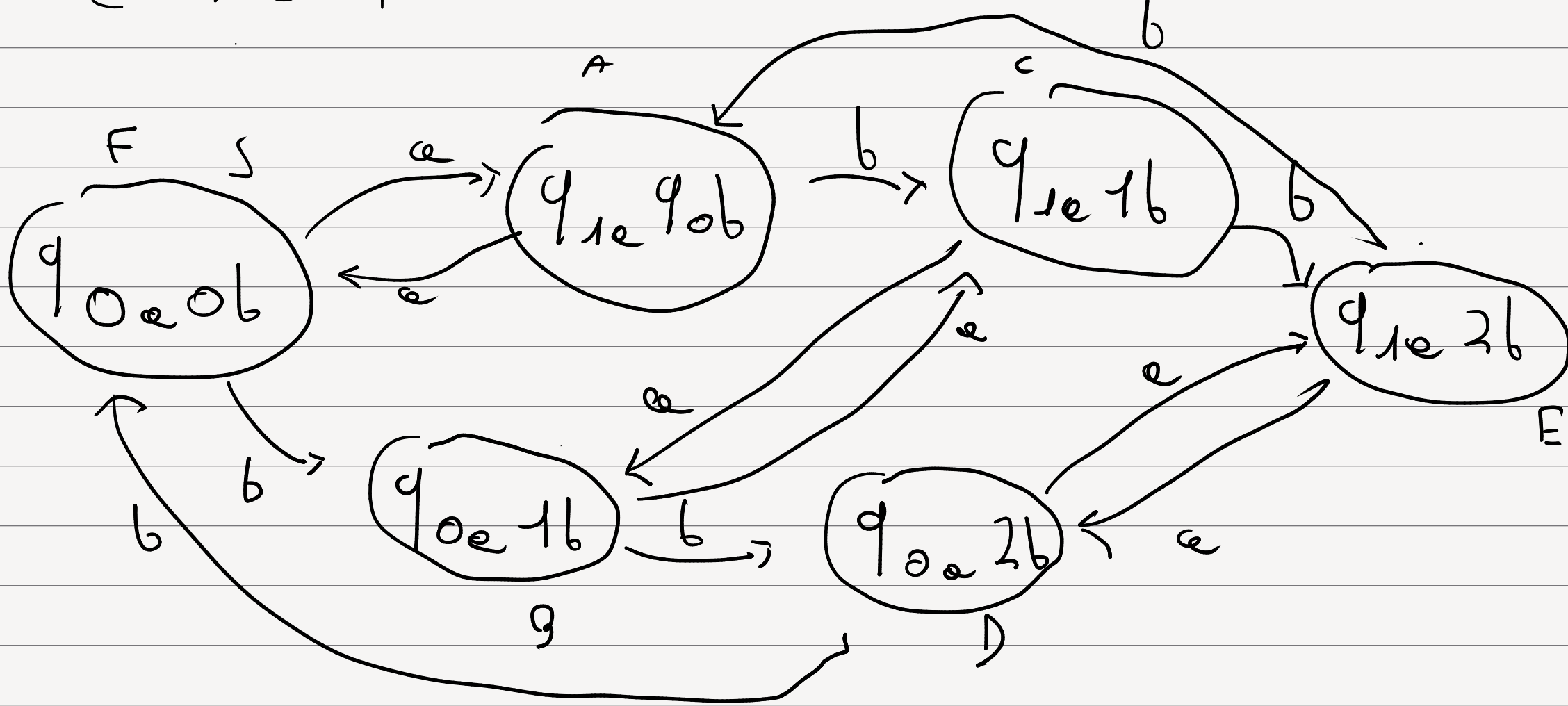
$$\delta(q_0, q_e) = e$$

Quindi:



~ ~ ~ ~

$$\{ w \in \{a, b\}^* \mid \#(a) = 2k, \#(b) = 3m, k, m \in \mathbb{N} \}$$



$$S \rightarrow aA \mid bB \mid \varepsilon$$

$$A \rightarrow aS \mid a \mid bC$$

$$B \rightarrow bD \mid aC$$

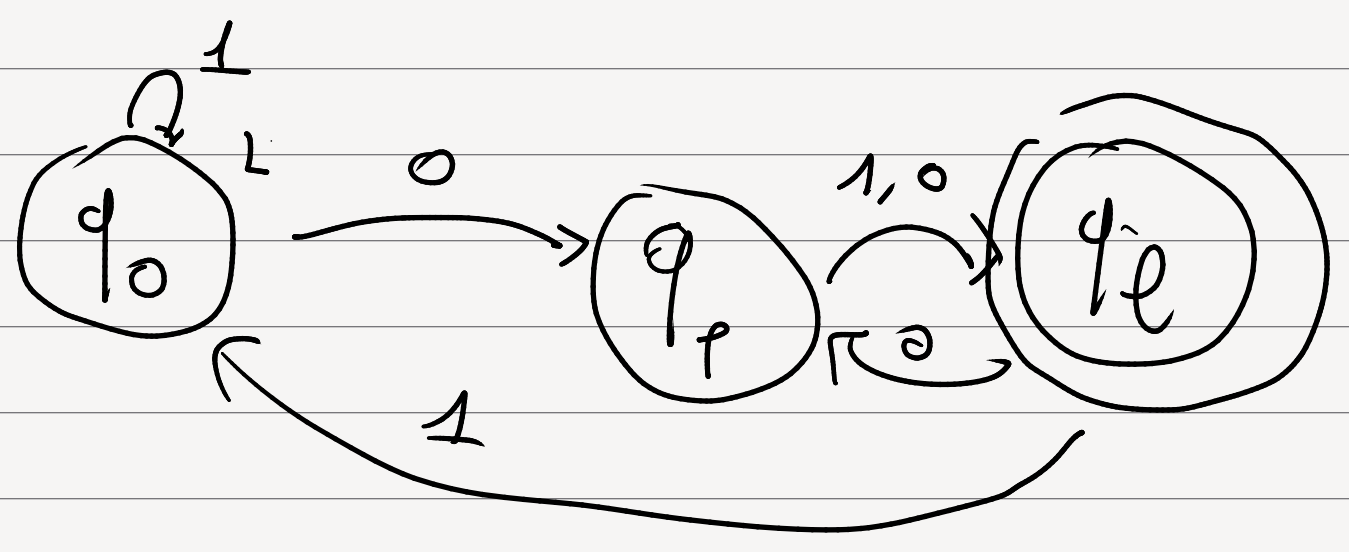
$$C \rightarrow bE \mid aB$$

$$D \rightarrow aE \mid bS \mid b$$

$$E \rightarrow aD \mid bA$$

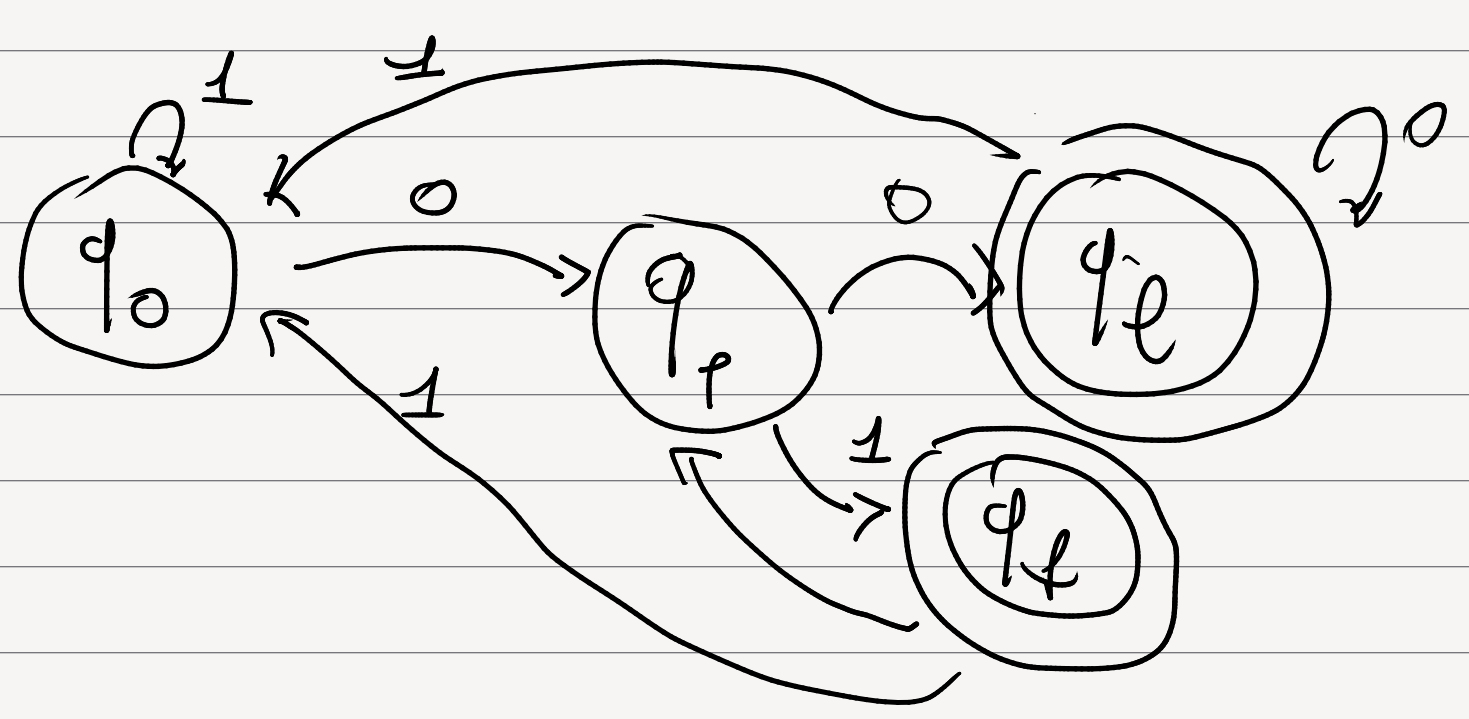
~ ~ ~ ~

$\{w \in \{0,1\}^* \mid w \text{ ha } 0 \text{ come penultimo simbolo}\}$



$S \rightarrow 1S \mid 0P$
 $P \rightarrow 1L \mid 0C \mid 1 \mid 0$
 $L \rightarrow 0P \mid 1S$

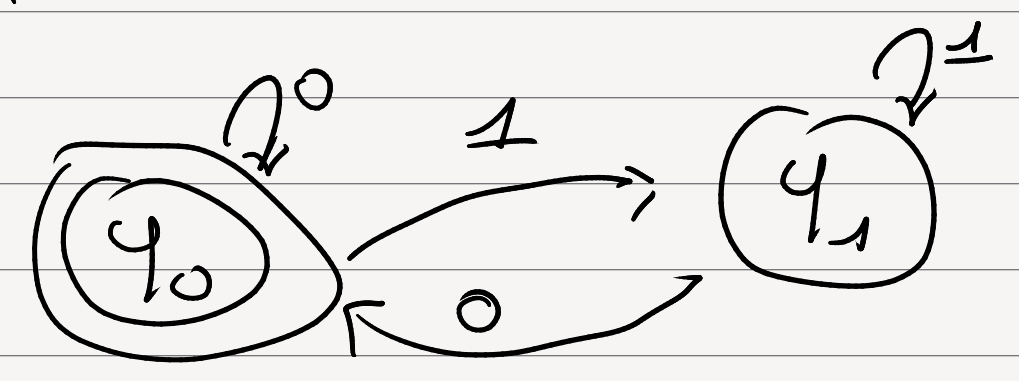
Questo non va bene in quanto stringhe formate da tutti 0 di lunghezza dispari, come 000 non vengono accettate. La versione corretta è la seguente.



~ ~ ~ ~

$\{w \in \{0,1\}^* \mid w \text{ divisibile per } 2\}$

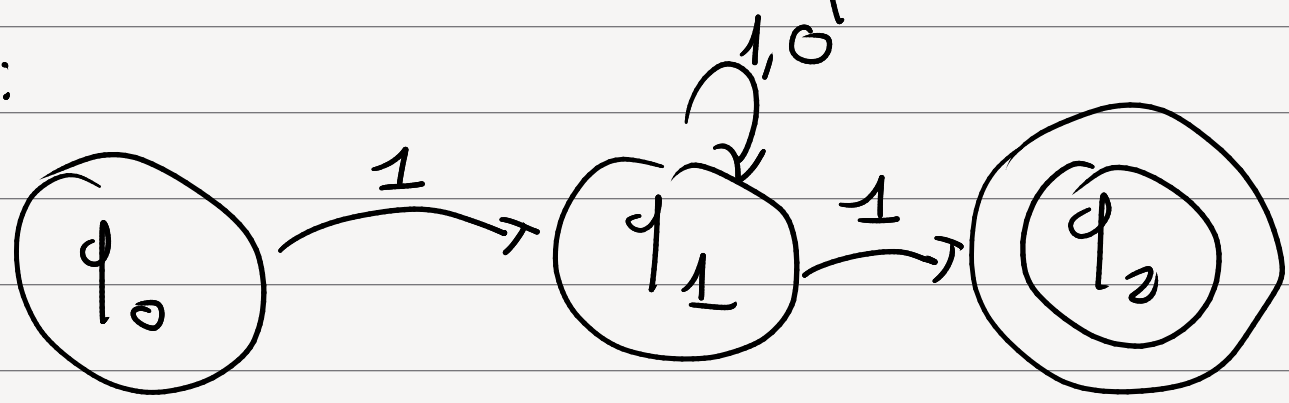
w è divisibile per 2 se è pari e le stringhe binarie pari terminano con 0.



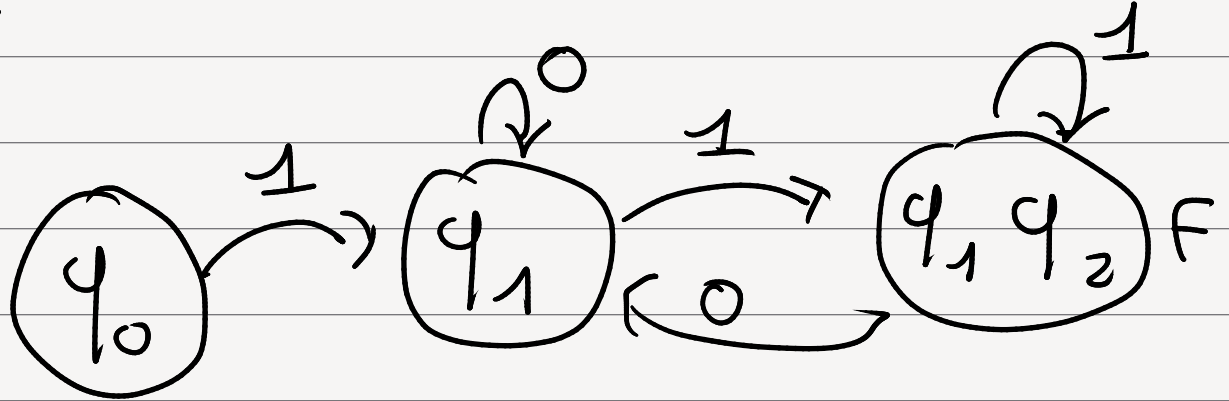
~ ~ ~ ~ ~

Scrivere DFA per $1(011)^*1$

NFA:



DFA:



~ ~ ~ ~ ~

DFA per $0(0110)^*0$

